

Performance Test

Test Run ID: 97771403-3514-4bf3-a8ea-1f715780e7d6

This chapter provides a summary of the test run metrics. The tables contain the aggregated values of the metrics for the entire test run.

Date	05 June 2026	Max VUs Tested	250
Executor	ramping-arrival-rate	Tool	k6

1. Executive Summary

The target endpoint for this performance test is the "https://quickpizza.grafana.com" URL. The testing type is a load test, utilizing a ramping arrival rate scenario to simulate increasing traffic. The load profile consists of a maximum of 100 Virtual Users (VUs) with a pre-allocated initial capacity of 10 VUs, ramping up to a peak rate of 250 requests per second (RPS) over a 30-second duration.

Stable Threshold	Degradation Onset	Critical Point	Peak Throughput
58-160 TPS	N/A	N/A	160 TPS

2. Performance Summary

Trends

metric	avg	max	med	min	p90	p95	p99
http_req_duration	260.90ms	1612.53ms	232.71ms	213.64ms	377.45ms	409.95ms	-
http_req_waiting	259.60ms	1541.62ms	232.37ms	212.92ms	375.84ms	409.08ms	-
http_req_connecting	10.58ms	292.22ms	0.00ms	0.00ms	0.00ms	0.00ms	-
iteration_duration	1283.49ms	2614.46ms	1234.80ms	1214.05ms	1408.52ms	1486.58ms	-

Counters

metric	count	rate
data_received	7.50 MB	245.93 kB/s
data_sent	284.22 kB	9.10 kB/s
http_reqs	2,197	70.36/s
iterations	2,197	70.36/s

Rates

metric	rate
checks_succeeded	100.00% (2197/2197)
checks_failed	0.00% (0/2197)
http_req_failed	0.00% (0/2197)

Gauges

metric	value
vus	30 (min: 30, max: 100)
vus_max	100 (min: 31, max: 100)

Checks

Check Name	Passed	Failed	Success Rate	Status
status is 200	2197	0	100.00%	Pass

3. Advanced Analysis

3.1 Error Analysis

- **Network/Connection errors:** None detected.
- **Application errors:** None detected.
- **Infrastructure errors:** None detected.
- **Warnings:** Insufficient VUs, reached 100 active VUs and cannot initialize more (logged at 2026-06-05T20:34:19+05:30).

3.2 Observation

The observed warning message "Insufficient VUs, reached 100 active VUs and cannot initialize more" indicates that the test reached its maximum VU (Virtual User) capacity of 100 and was unable to initialize more. This is likely due to the test's ramping arrival rate strategy, which attempts to gradually increase the number of VUs over time. However, in this case, the test was unable to scale beyond 100 VUs, resulting in a warning message.

The lack of any detected errors, warnings, or failed checks suggests that the system under test (SUT) was able to handle the load imposed by the test without any significant issues. However, the warning message indicates that the test may not have been able to fully stress the system, potentially limiting the effectiveness of the performance test.

3.3 Final Recommendations

1. **Advanced Connection Management:** Implement a connection pooling mechanism to reduce the overhead of establishing new connections and improve the overall performance of the system.
2. **Session Persistence sticky affinity/Redis cache:** Implement a session persistence mechanism to reduce the overhead of session initialization and improve the overall performance of the system.
3. **Gateway resilience Istio outlier circuit breaker:** Implement an outlier detection mechanism to detect and prevent cascading failures in the system.
4. **Observability tracing and standardized log levels:** Implement a tracing mechanism to provide detailed insights into the system's behavior and improve the overall observability of the system.
5. **Database read/write splitting and PgBouncer connection pooling:** Implement a read/write splitting mechanism to reduce the load on the database and improve the overall performance of the system.

3.4 Conclusion

The performance test results indicate that the system under test was able to handle the load imposed by the test without any significant issues. However, the warning message "Insufficient VUs, reached 100 active VUs and cannot initialize more" suggests that the test may not have been able to fully stress the system. To improve the effectiveness of the performance test, it is recommended to implement the above-mentioned recommendations to improve the overall performance and resilience of the system.